

A case-control study on seaweed consumption and the risk of breast cancer.

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Gim (*Porphyra* sp.) and miyeok (*Undaria pinnatifida*) are the seaweeds most consumed by Koreans. We investigated the association between the intake of gim and miyeok and the risk of breast cancer in a case-control study. Cases were 362 women aged 30-65 years old, who were histologically confirmed to have breast cancer. Controls visiting the same hospital were matched to cases according to their age (sd 2 years) and menopausal status. Food intake was estimated by the quantitative FFQ with 121 items, including gim and miyeok. Conditional logistic regression analysis was used to obtain the OR and corresponding 95 % CI. The average intake and consumption frequency of gim in cases were lower than in controls. The daily intake of gim was inversely associated with the risk of breast cancer (5th v. 1st quintile, OR, 0.48; 95 % CI, 0.27, 0.86; P for trend, 0.026) after adjustment for potential confounders. After stratification analysis was performed according to menopausal status, premenopausal women (5th v. 1st quintile, OR, 0.44; 95 % CI, 0.24, 0.80; P for trend, 0.007) and postmenopausal women (5th v. 1st quintile, OR, 0.32; 95 % CI, 0.13, 0.80; P for trend, 0.06) showed similar inverse associations between gim intake and the risk of breast cancer after an adjustment for potential confounders except dietary factors. Miyeok consumption did not have any significant associations with breast cancer. These results suggest that high intake of gim may decrease the risk of breast cancer.